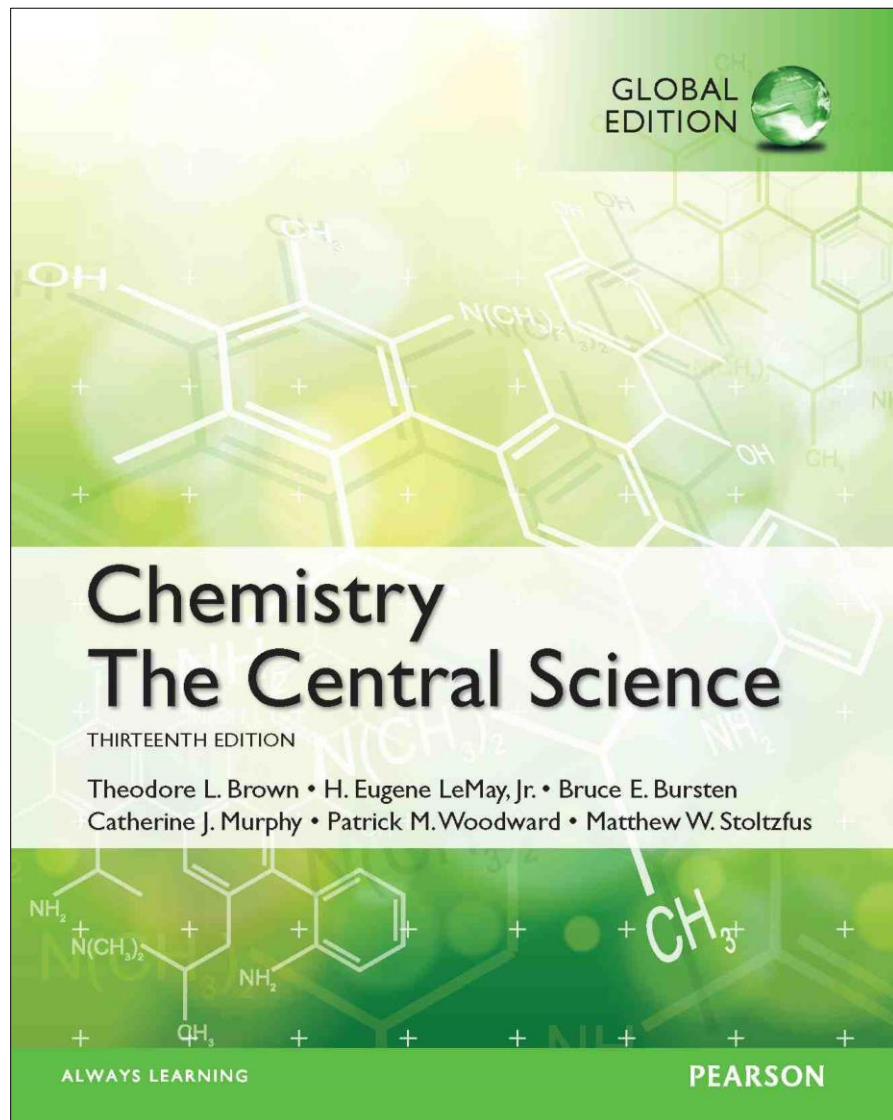




Lecture Presentation

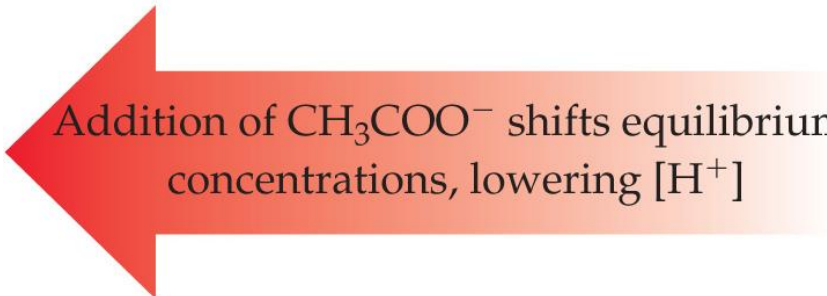
Chapter 17

Additional Aspects of Aqueous Equilibria

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Effect of Acetate on the Acetic Acid Equilibrium

- Acetic acid is a weak acid:
 - $\text{CH}_3\text{COOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{CH}_3\text{COO}^-(aq)$
- Sodium acetate is a strong electrolyte:
 - $\text{NaCH}_3\text{COO}(aq) \rightarrow \text{Na}^+(aq) + \text{CH}_3\text{COO}^-(aq)$
- The presence of acetate from sodium acetate in an acetic acid solution will shift the equilibrium, according to LeChâtelier's Principle:
 - $\text{CH}_3\text{COOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{CH}_3\text{COO}^-(aq)$



Addition of CH_3COO^- shifts equilibrium concentrations, lowering $[\text{H}^+]$



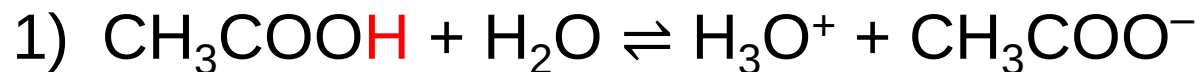
Aqueous
Equilibria

The Common-Ion Effect

- “Whenever a weak electrolyte and a strong electrolyte containing a common ion are together in solution, the weak electrolyte ionizes less than it would if it were alone in solution.”
- This affects acid–base equilibria.
- We will also see (later in the chapter) how it affects solubility.



- Calculate the pH of a 0.30 *M* solution of acetic acid, CH₃COOH, at 25 °C.



2) $K_a = [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] = 1.8 \times 10^{-5}$



- Calculate the pH of a 0.30 M solution of acetic acid, CH₃COOH, at 25 °C.



2) $K_a = [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$

3)

	CH ₃ COOH (M)	H ₃ O ⁺ (M)	CH ₃ COO ⁻ (M)
Initial Concentration (M)	0.30	0	0
Change in Concentration (M)	-x	+x	+x
Equilibrium Concentration (M)	0.30 - x	x	x



$$\begin{aligned} 4) \quad K_a &= [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] \\ &= (x)(x) / (0.30 - x) \end{aligned}$$

If we assume that $x \ll 0.30$ (**if smaller than 5% or 1/20**), then $(0.30 - x) \approx 0.30$. The problem becomes easier, since we don't have to use the quadratic formula to solve it.

$$K_a = 1.8 \times 10^{-5} = x^2 / 0.30$$

$$\text{so } x = 2.3 \times 10^{-3} \text{ (indeed, } x \ll 0.30 \text{)}$$

$$x = [\text{H}_3\text{O}^+], \text{ so pH} = -\log(2.3 \times 10^{-3}) = \mathbf{2.64}$$



An Acid–Base Example

- What is the pH of a solution made by adding 0.30 mol of acetic acid and 0.30 mol of sodium acetate to enough water to make 1.0 L of solution?

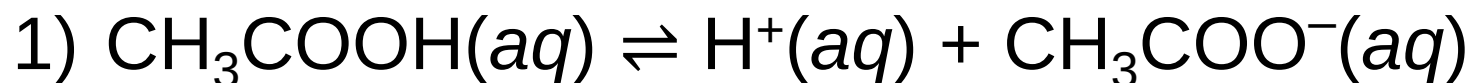


2) $K_a = [\text{H}^+][\text{CH}_3\text{COO}^-]/[\text{CH}_3\text{COOH}] = 1.8 \times 10^{-5}$



An Acid–Base Example

- What is the pH of a solution made by adding 0.30 mol of acetic acid and 0.30 mol of sodium acetate to enough water to make 1.0 L of solution?



2) $K_a = [\text{H}^+][\text{CH}_3\text{COO}^-]/[\text{CH}_3\text{COOH}] = 1.8 \times 10^{-5}$

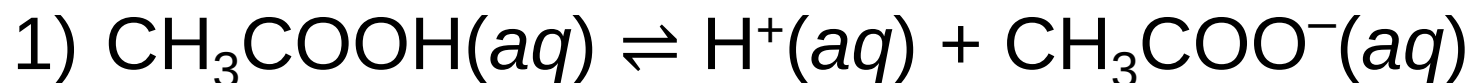
3)

	$[\text{CH}_3\text{COOH}] (M)$	$[\text{H}^+] (M)$	$[\text{CH}_3\text{COO}^-] (M)$
Initial Concentration (M)	0.30	0	0.30
Change in Concentration (M)	-x	+x	+x
Equilibrium Concentration (M)	$0.30 - x$	x	$0.30 + x$



An Acid–Base Example

- What is the pH of a solution made by adding 0.30 mol of acetic acid and 0.30 mol of sodium acetate to enough water to make 1.0 L of solution?



2) $K_a = [\text{H}^+][\text{CH}_3\text{COO}^-]/[\text{CH}_3\text{COOH}] = 1.8 \times 10^{-5}$

3)

	$[\text{CH}_3\text{COOH}] (M)$	$[\text{H}^+] (M)$	$[\text{CH}_3\text{COO}^-] (M)$
Initial Concentration (M)	0.30	0	0.30
Change in Concentration (M)	-x	+x	+x
Equilibrium Concentration (M)	$0.30 - x$	x	$0.30 + x$



$$4) \quad 1.8 \times 10^{-5} = (x)(0.30 + x)/(0.30 - x)$$

Assume that adding or subtracting x from 0.30 will not change 0.30 enough to matter and the equation becomes

$$[\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] = [\text{CH}_3\text{COO}^-]_0 / [\text{CH}_3\text{COOH}]_0$$

$$1.8 \times 10^{-5} = (x)(0.30)/(0.30)$$

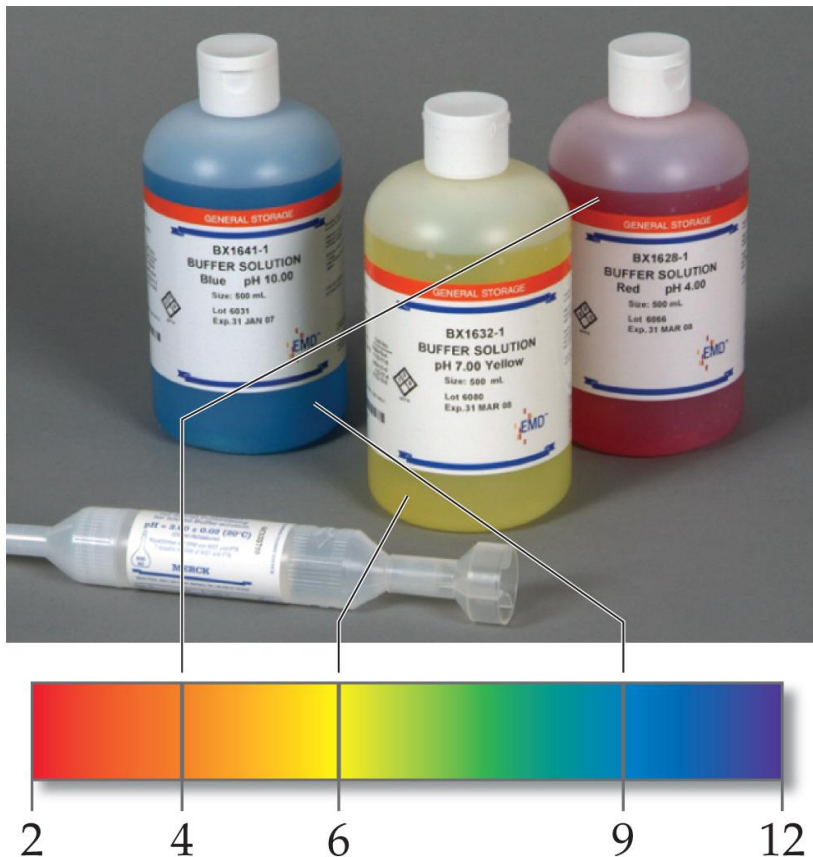
which results in: $x = 1.8 \times 10^{-5} = [\text{H}^+]$

$$\text{So: } \text{pH} = -\log[\text{H}^+] = 4.74$$



Buffers

- Solutions of a weak conjugate acid–base pair that resist drastic changes in pH are called **buffers**.
- These solutions contain relatively **high concentrations** ($10^{-3} M$ or more) of both the acid and base. Their concentrations are **approximately equal**.

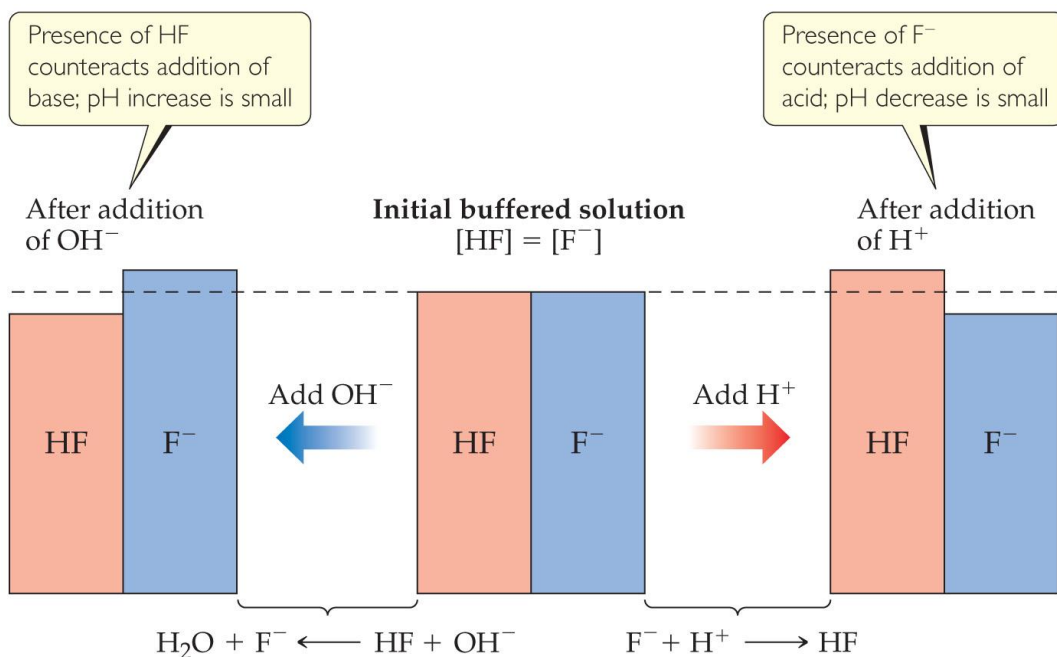


Ways to Make a Buffer

- 1) Mix a weak acid and a salt of its conjugate base or a weak base and a salt of its conjugate acid.
- 2) Add strong acid and partially neutralize a weak base or add strong base and partially neutralize a weak acid.



How a Buffer Works



- Adding a **small** amount of acid or base only slightly neutralizes one component of the buffer, so the pH doesn't change much.

Calculating the pH of a Buffer

- For a weak acid: $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$
- Take $-\log$ of both sides:
$$-\log K_a = -\log[\text{H}^+] + -\log([\text{A}^-]/[\text{HA}])$$
- Rearrange:
$$-\log[\text{H}^+] = -\log K_a + \log([\text{A}^-]/[\text{HA}])$$
- Which is:
$$\text{pH} = \text{p}K_a + \log([\text{A}^-]/[\text{HA}])$$
- This equation is known as the **Henderson–Hasselbalch equation**. This applies only to buffers.



Henderson–Hasselbalch Equation

What is the pH of a buffer that is 0.12 *M* in lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and 0.10 *M* in sodium lactate? K_a for lactic acid is 1.4×10^{-4} .

$$\begin{aligned}\text{pH} &= \text{p}K_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) \\ &= -\log(1.4 \times 10^{-4}) + \log[(0.10 \text{ M})/(0.12 \text{ M})] \\ &= 3.85 + (-0.08) = 3.77\end{aligned}$$



Buffer Capacity

- The amount of acid or base the buffer can neutralize before the pH begins to change to an **appreciable** degree
- Using the Henderson–Hasselbalch equation, pH will be the same for a conjugate acid–base pair of 1 *M* each or 0.1 *M* each; however, the buffer which is 1 *M* can neutralize more acid or base before the pH changes.

$$\text{pH} = \text{p}K_a + \log([A^-]/[HA])$$



pH Range

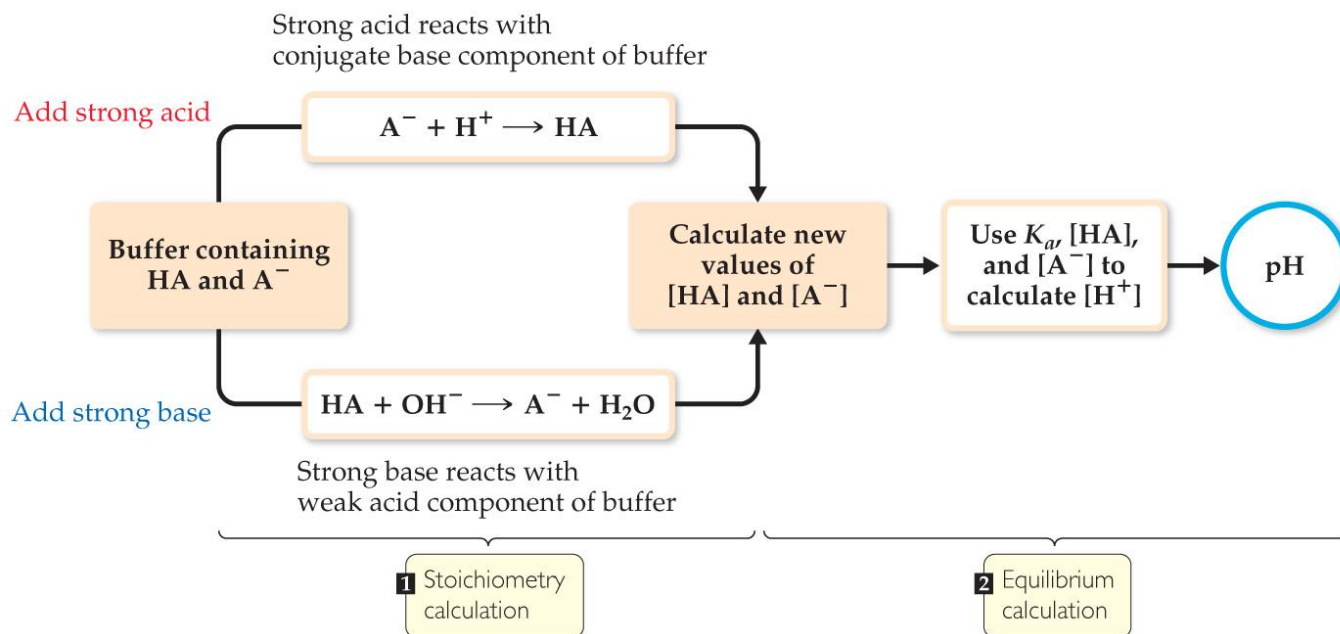
- The range of pH values over which a buffer system works effectively
- Optimal pH: where $\text{pH} = \text{p}K_a$ ($[\text{HA}] = [\text{A}^-]$)
- If one concentration is more than 10 times the other, the buffering action is poor; this means that the pH range of a buffer is usually ± 1 pH unit from $\text{p}K_a$.

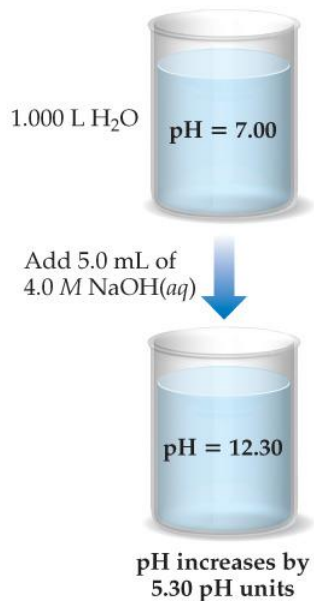
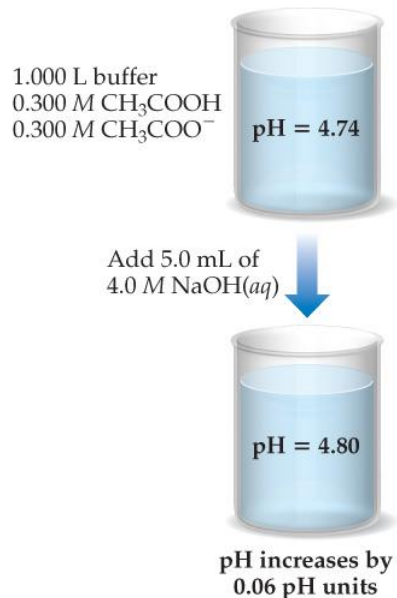
between $(\text{p}K_a - 1)$ and $(\text{p}K_a + 1)$



Addition of a Strong Acid or a Strong Base to a Buffer

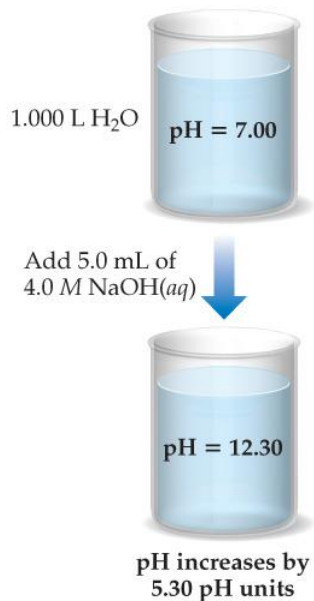
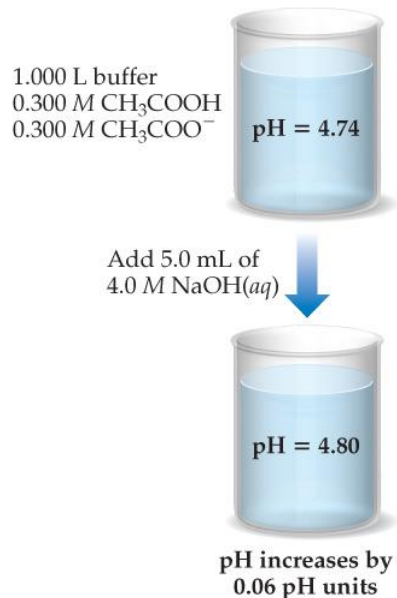
- 1) Adding the **strong** acid or base is a neutralization reaction; calculate like stoichiometry problem to find $[HA]$ and $[A^-]$ when *all* of the added acid or base reacts.
- 2) Use the Henderson–Hasselbalch equation to find pH.





- A buffer is made by adding 0.300 mol $\text{HC}_2\text{H}_3\text{O}_2$ and 0.300 mol $\text{NaC}_2\text{H}_3\text{O}_2$ to enough water to make 1.00 L of solution. Calculate the pH after **0.020 mol of NaOH** is added.





- A buffer is made by adding 0.300 mol $\text{HC}_2\text{H}_3\text{O}_2$ and 0.300 mol $\text{NaC}_2\text{H}_3\text{O}_2$ to enough water to make 1.00 L of solution. Calculate the pH after **0.020 mol of NaOH** is added.

1)

	$[\text{CH}_3\text{COOH}]$	$[\text{OH}^-]$	$[\text{CH}_3\text{COO}^-]$
Before reaction (mol)	0.300	0.020	0.300
Change (mol)	-0.020	-0.020	+0.020
After reaction (mol)	0.280	0	0.320



2) Use the Henderson–Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log([A^-]/[HA])$$

Since this is a buffer, the volume for each concentration is the same, so the ratio of molarity can be calculated using a ratio of moles.

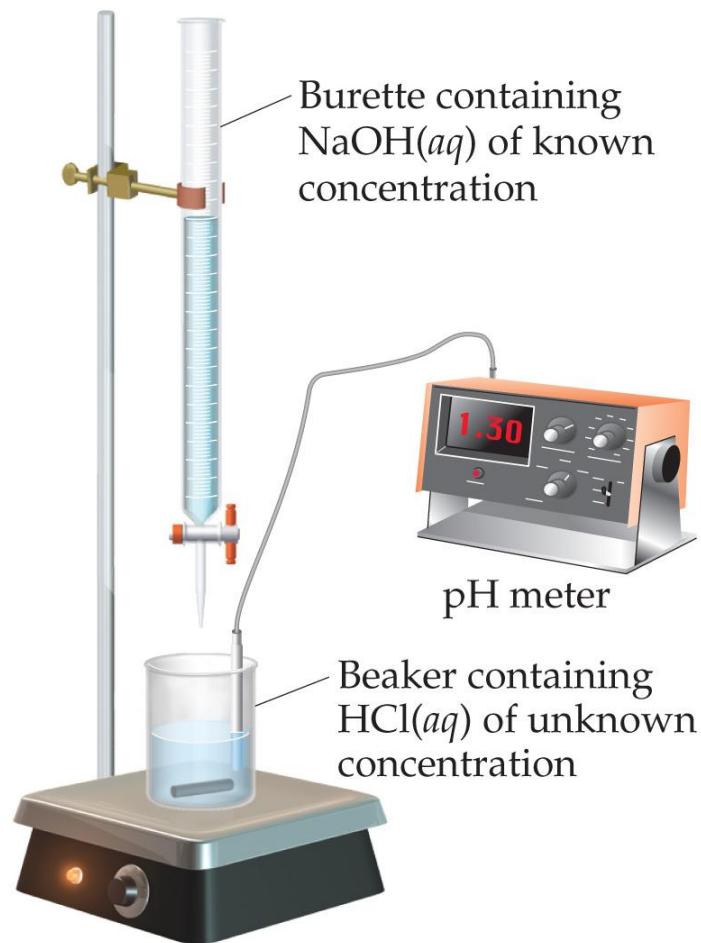
$$\text{pH} = \text{p}K_a + \log(n_{A^-}/n_{HA})$$

$$\text{pH} = 4.74 + \log(0.320/0.280) = 4.80$$



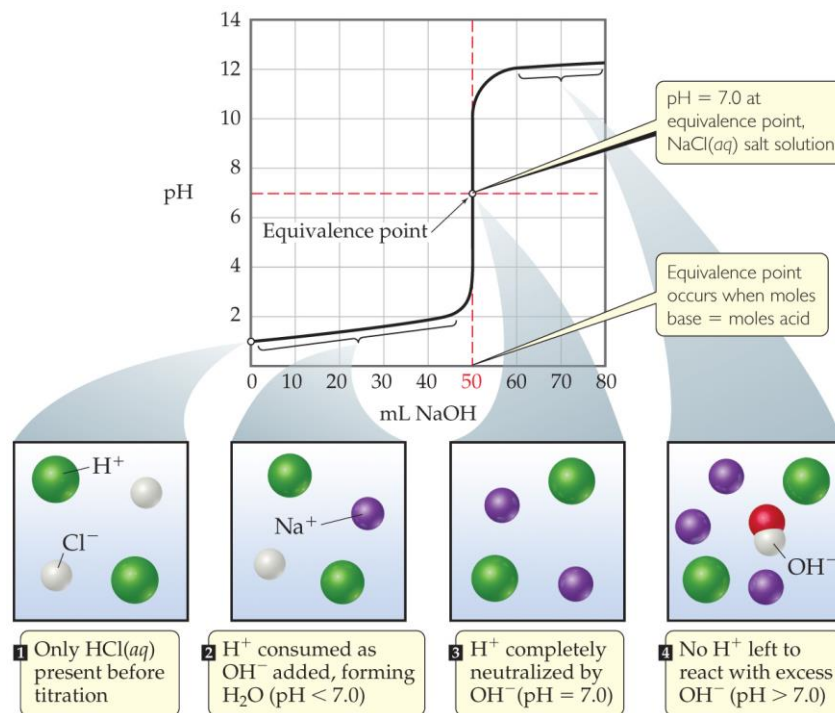
Titration

- In this technique, an acid (or base) solution of **known concentration** is slowly added to a base (or acid) solution of **unknown concentration**.
- A pH meter or indicators are used to determine when the solution has reached the **equivalence point**:
The amount of acid equals that of base.



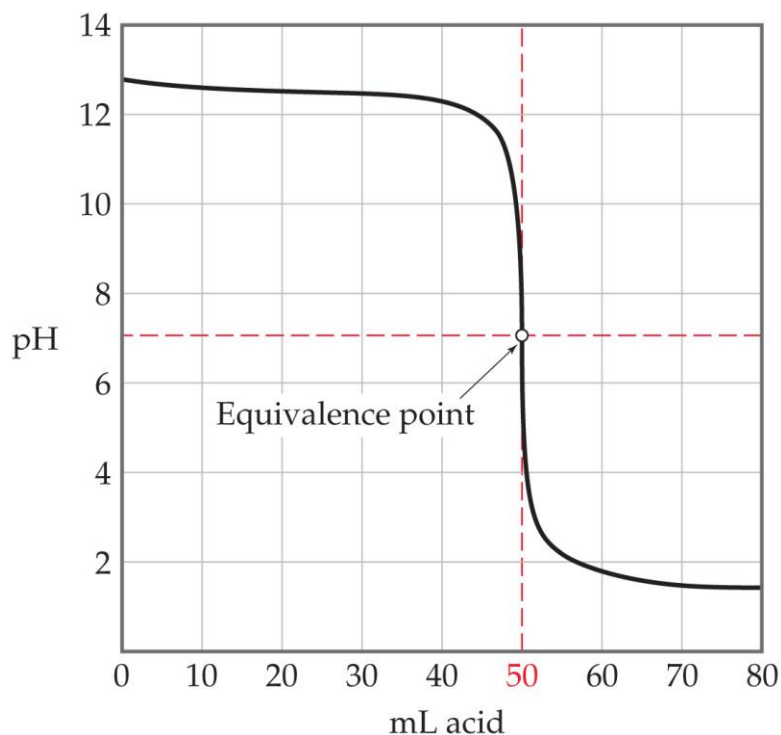
Titration of a **Strong** Acid with a **Strong** Base

- From the start of the titration to near the equivalence point, the pH goes up **slowly**.
- Just before (and after) the equivalence point, the pH rises **rapidly**.
- At the equivalence point, $\text{pH} = 7$.
- As more base is added, the pH again levels off.



Aqueous
Equilibria

Titration of a **Strong** Base with a **Strong** Acid



- It looks like you “flipped over” the strong acid being titrated by a strong base.
- Start with a high pH (basic solution); the pH = 7 at the equivalence point; low pH to end.

SAMPLE

EXERCISE 17.7

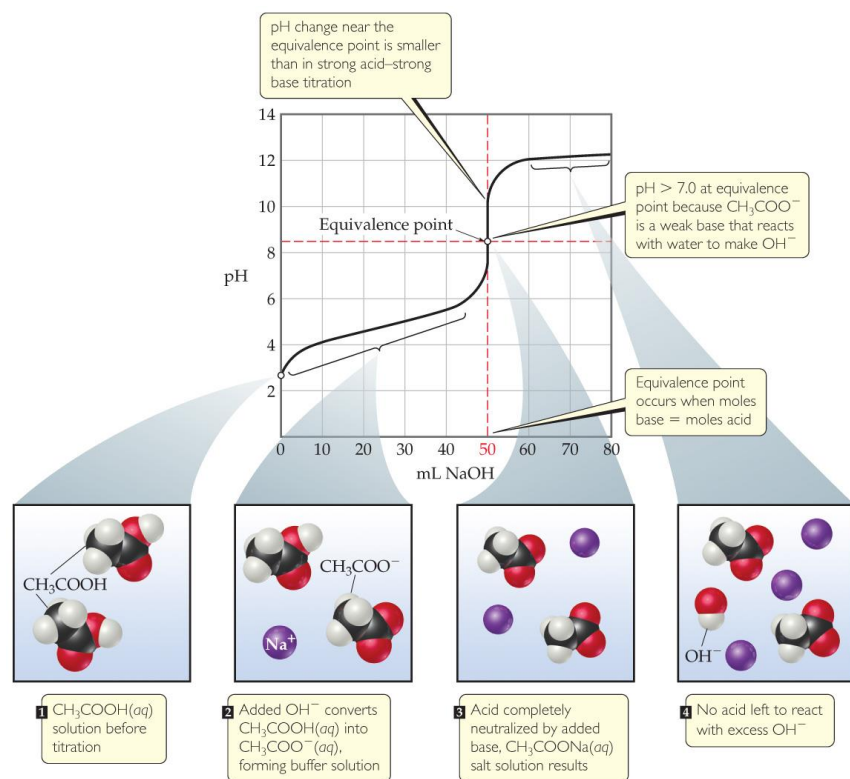
Calculations for a Strong Acid–Strong Base Titration

Calculate the pH when (a) 49.0 mL and (b) 51.0 mL of 0.100 *M* NaOH solution have been added to 50.0 mL of 0.100 *M* HCl solution.



Titration of a **Weak** Acid with a **Strong** Base

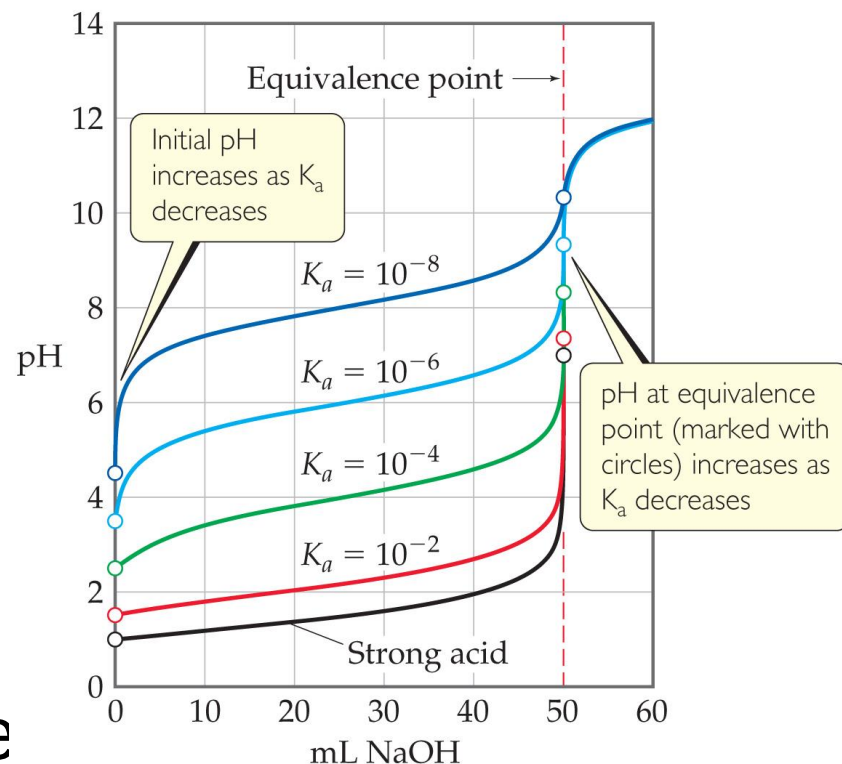
- Use K_a to find initial pH.
- Find the pH in the “buffer region” using stoichiometry followed by the Henderson–Hasselbalch equation.
- At the equivalence point the pH is >7 . Use the conjugate base of the weak acid to determine the pH.
- As more base is added, the pH levels off. This is exactly the same as for strong acids.



Aqueous
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Ways That a Weak Acid Titration Differs from a Strong Acid Titration

- 1) A solution of weak acid has a higher initial pH than a strong acid.
- 2) The pH change near the equivalence point is smaller for a weak acid. (This is at least partly due to the buffer region.)
- 3) The pH at the equivalence point is greater than 7 for a weak acid.



SAMPLE

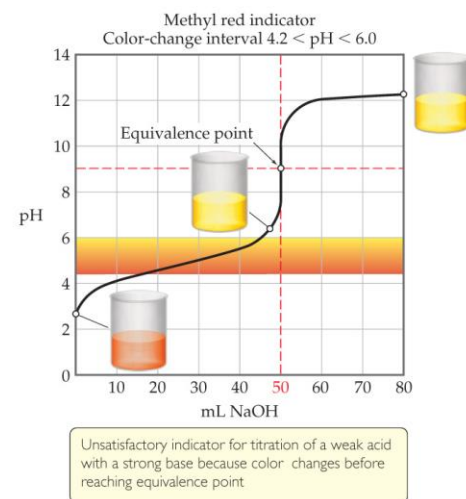
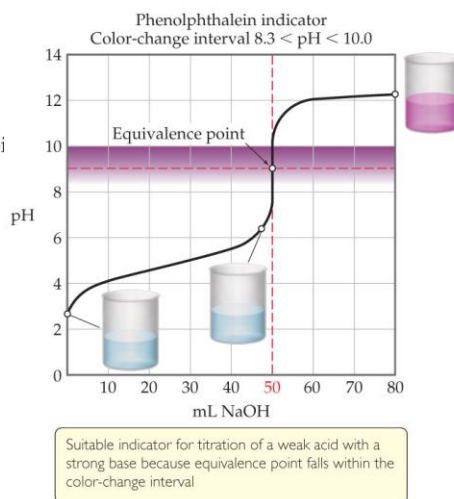
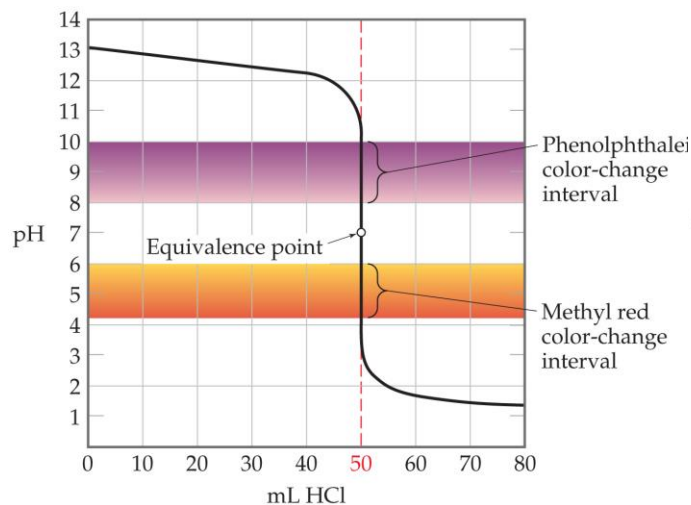
EXERCISE 17.8 Calculations for a Weak Acid–Strong Base Titration

Calculate the pH of the solution formed when 45.0 mL of 0.100 *M* NaOH is added to 50.0 mL of 0.100 *M* CH₃COOH ($K_a = 1.8 \times 10^{-5}$).

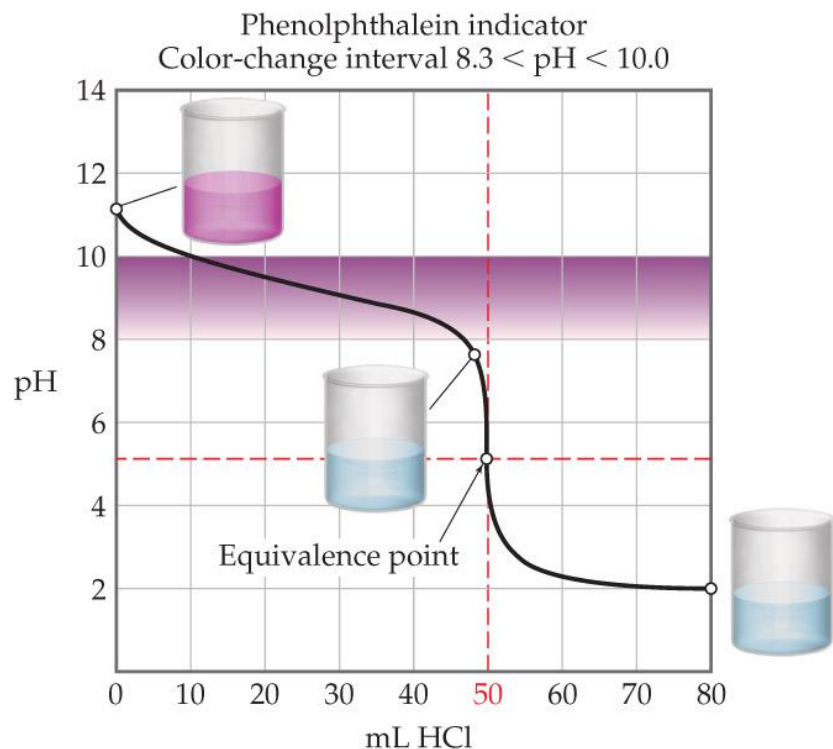


Use of Indicators

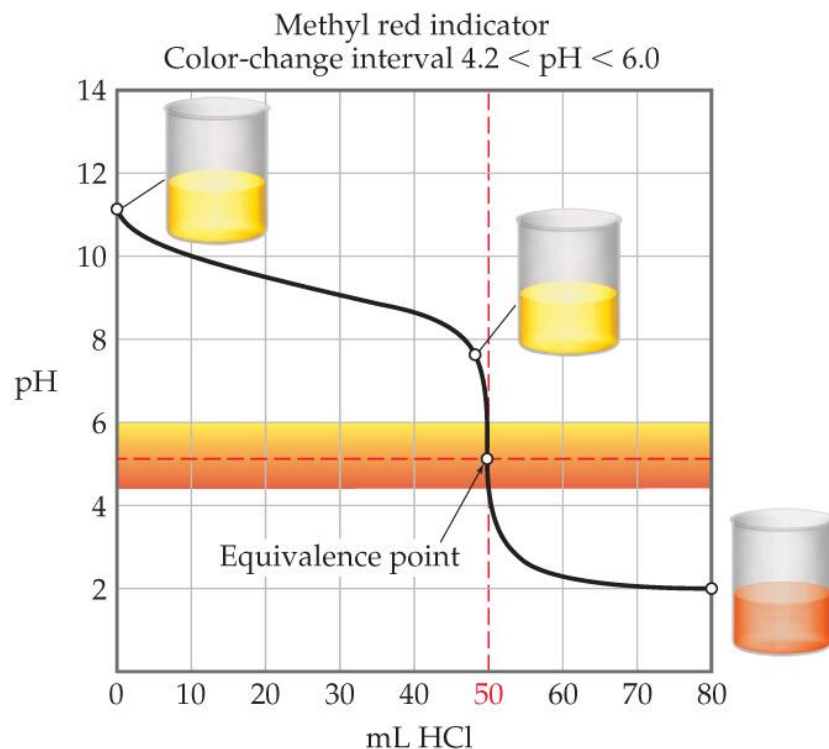
- Indicators are weak acids that have a different color than their conjugate base form.
- Each indicator has its own pH range over which it changes color.
- An indicator can be used to find the equivalence point in a titration as long as it changes color in the small volume change region where the pH rapidly changes.



Indicator Choice Can Be Critical!



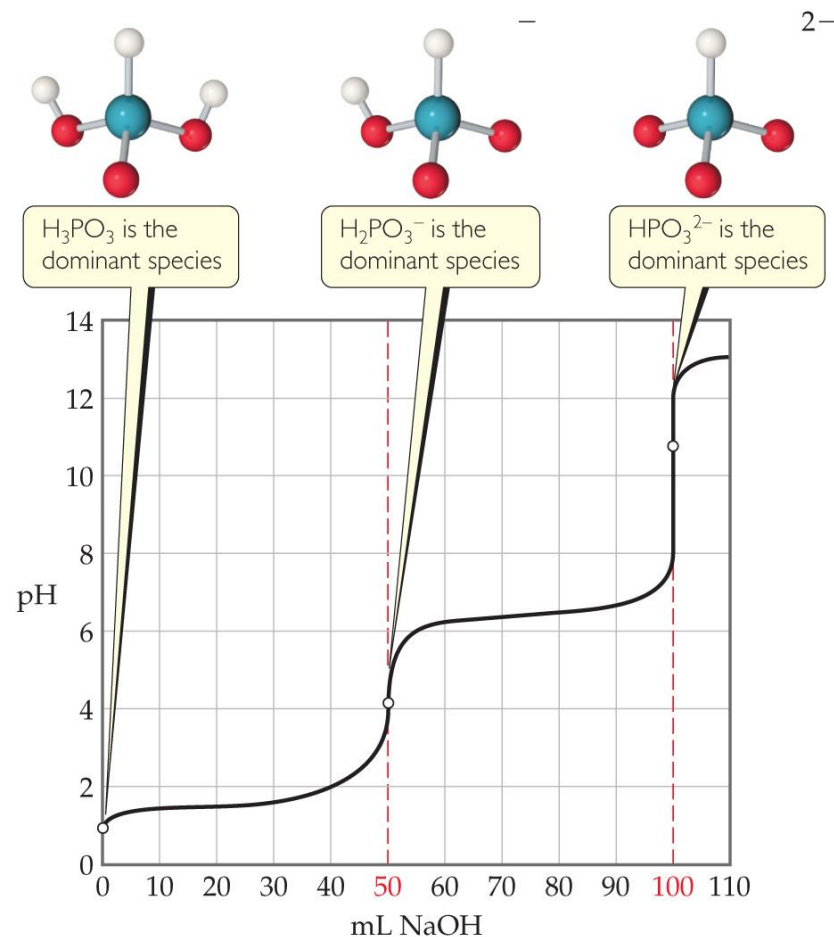
Unsatisfactory indicator for titration of a weak base with a strong acid because color changes before reaching equivalence point



Suitable indicator for titration of a weak base with a strong acid because equivalence point falls within the color-change interval

Titrations of Polyprotic Acids

- When a polyprotic acid is titrated with a base, there is an equivalence point for each dissociation.
- Using the Henderson–Hasselbalch equation, we can see that *half* way to each equivalence point gives us the pK_a for that step.



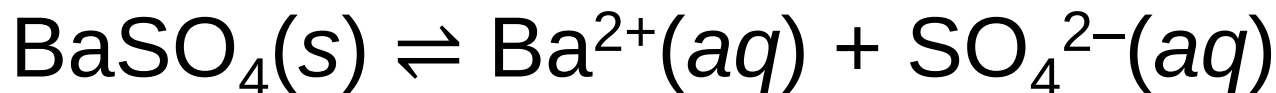
Solubility Equilibria

- Because ionic compounds are strong electrolytes, they dissociate completely to the extent that they dissolve.
- When an equilibrium equation is written, the solid is the reactant and the ions in solution are the products.
- The equilibrium constant expression is called the **solubility-product constant**. It is represented as K_{sp} .



Solubility Product

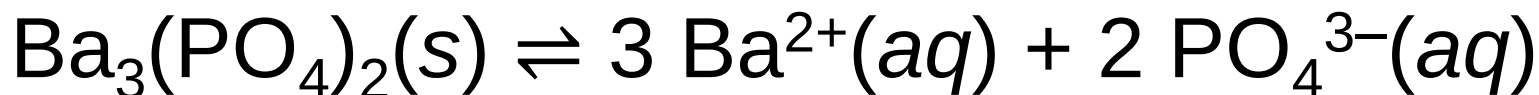
- For example:



- The equilibrium constant expression is

$$K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}]$$

- Another example:



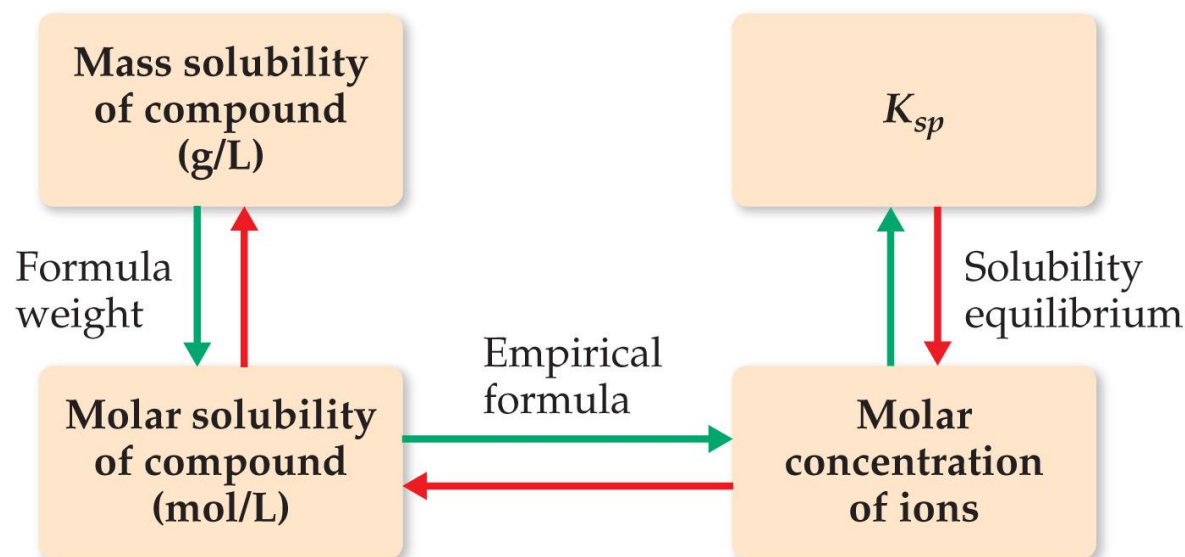
- The equilibrium constant expression is

$$K_{sp} = [\text{Ba}^{2+}]^3[\text{PO}_4^{3-}]^2$$



Solubility vs. Solubility Product

- K_{sp} is *not* the same as solubility.
- Solubility is the quantity of a substance that dissolves to form a saturated solution
- Common units for solubility:
 - Grams per liter (g/L)
 - Moles per liter (mol/L)



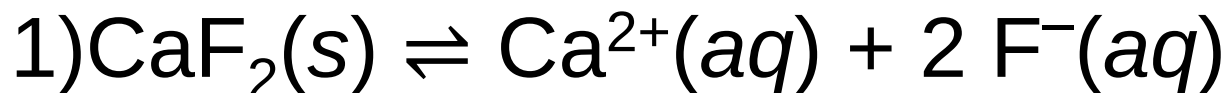
Calculating Solubility from K_{sp}

- The K_{sp} for CaF_2 is 3.9×10^{-11} at 25°C . What is its molar solubility?
- Follow the same format as before:
 - 1) $\text{CaF}_2(s) \rightleftharpoons \text{Ca}^{2+}(aq) + 2 \text{F}^{-}(aq)$
 - 2) $K_{sp} = [\text{Ca}^{2+}][\text{F}^{-}]^2 = 3.9 \times 10^{-11}$



Calculating Solubility from K_{sp}

- The K_{sp} for CaF_2 is 3.9×10^{-11} at 25°C . What is its molar solubility?
- Follow the same format as before:



2) $K_{sp} = [\text{Ca}^{2+}][\text{F}^{-}]^2 = 3.9 \times 10^{-11}$

3)

	$\text{CaF}_2(\text{s})$	$[\text{Ca}^{2+}](\text{M})$	$[\text{F}^{-}](\text{M})$
Initial concentration (M)	---	0	0
Change in concentration (M)	---	+x	+2x
Equilibrium concentration (M)	---	x	2x



Example (completed)

- 4) Solve: Substitute the equilibrium concentration values from the table into the solubility-product equation:

$$3.9 \times 10^{-11} = (x) (2x)^2 = 4x^3$$

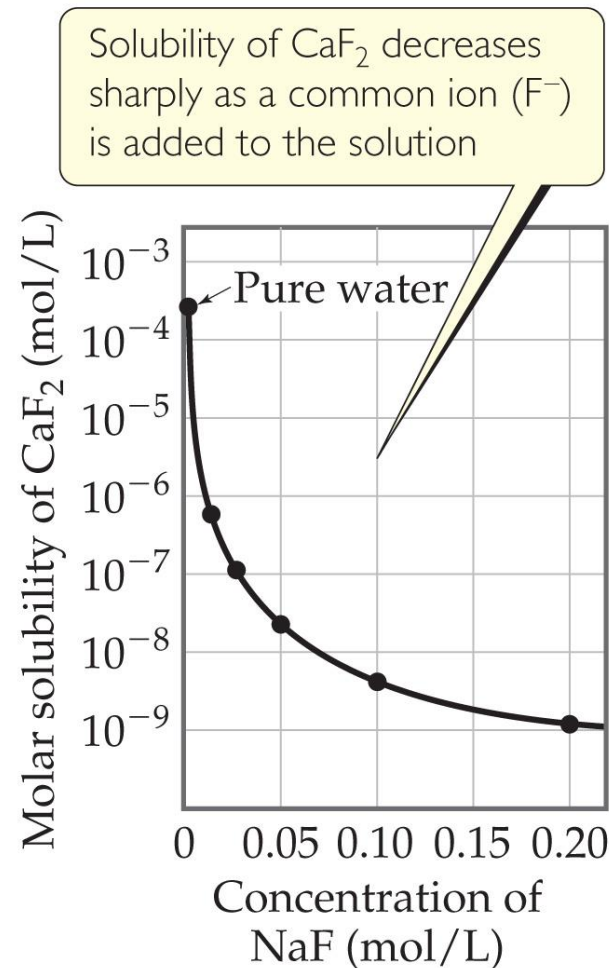
$$x = 2.1 \times 10^{-4} M$$

(If you want the answer in g/L, multiply by molar mass; this would give 0.016 g/L.)



Factors Affecting Solubility

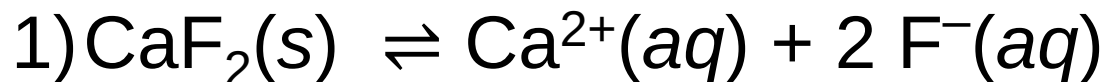
- The **Common-Ion** Effect
 - If one of the ions in a solution equilibrium is already dissolved in the solution, the solubility of the salt will decrease.
 - If either calcium ions or fluoride ions are present, then calcium fluoride will be less soluble.



Aqueous
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Calculating Solubility with a **Common Ion**

- What is the molar solubility of CaF_2 in $0.010\text{ M Ca(NO}_3)_2$?
- Follow the same format as before:

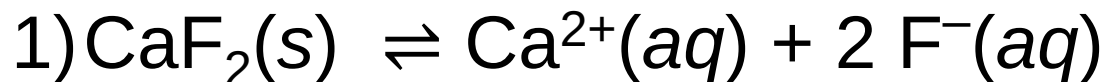


2) $K_{sp} = [\text{Ca}^{2+}][\text{F}^{-}]^2 = 3.9 \times 10^{-11}$



Calculating Solubility with a **Common Ion**

- What is the molar solubility of CaF_2 in $0.010\text{ M Ca(NO}_3)_2$?
- Follow the same format as before:



2) $K_{sp} = [\text{Ca}^{2+}][\text{F}^{-}]^2 = 3.9 \times 10^{-11}$

3)

	$\text{CaF}_2(\text{s})$	$[\text{Ca}^{2+}](\text{M})$	$[\text{F}^{-}](\text{M})$
Initial concentration (M)	---	0.010	0
Change in concentration (M)	---	+x	+2x
Equilibrium concentration (M)	---	$0.010 + x$	$2x$



Example (completed)

- 4) Solve: Substitute the equilibrium concentration values from the table into the solubility-product equation:

$$3.9 \times 10^{-11} = (0.010 + x)(2x)^2$$

(We assume that $x \ll 0.010$, so that $0.010 + x = 0.010$!)

$$3.9 \times 10^{-11} = (0.010)(2x)^2$$

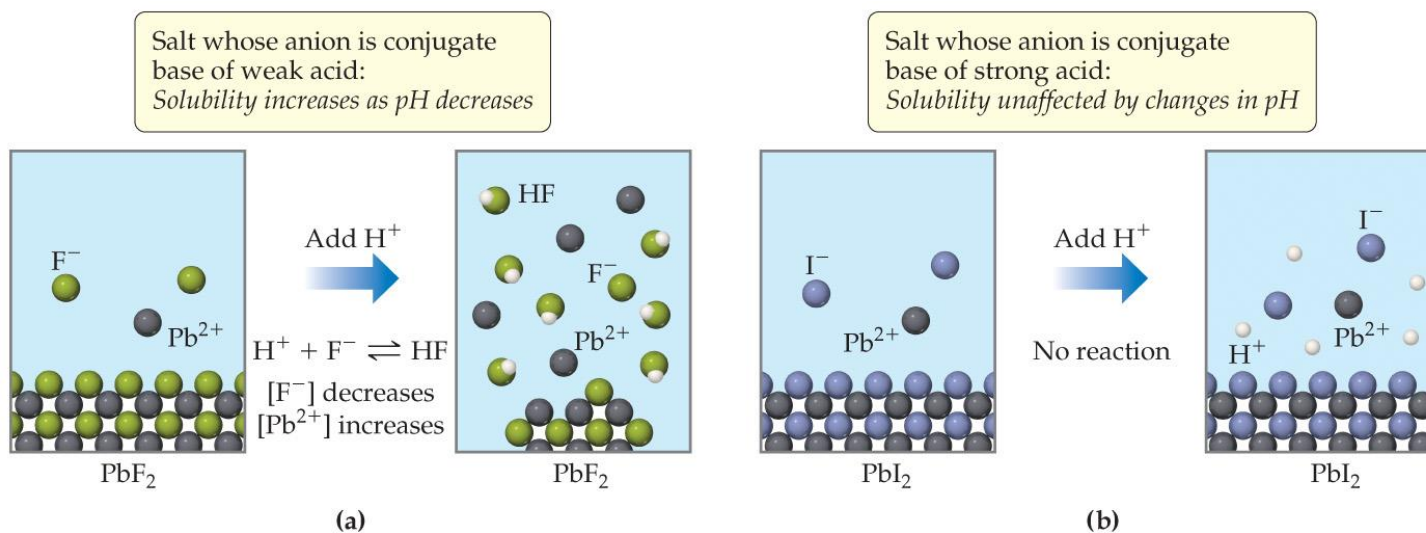
$$x = 3.1 \times 10^{-5} M$$

VS $x = 2.1 \times 10^{-4} M$



Factors Affecting Solubility

- **pH:** If a substance has a basic anion, it will be more soluble in an acidic solution.
- Remember that buffers control pH. When a buffer is used, there is *no change* in concentration of hydroxide ion!



Complex Ion Formation

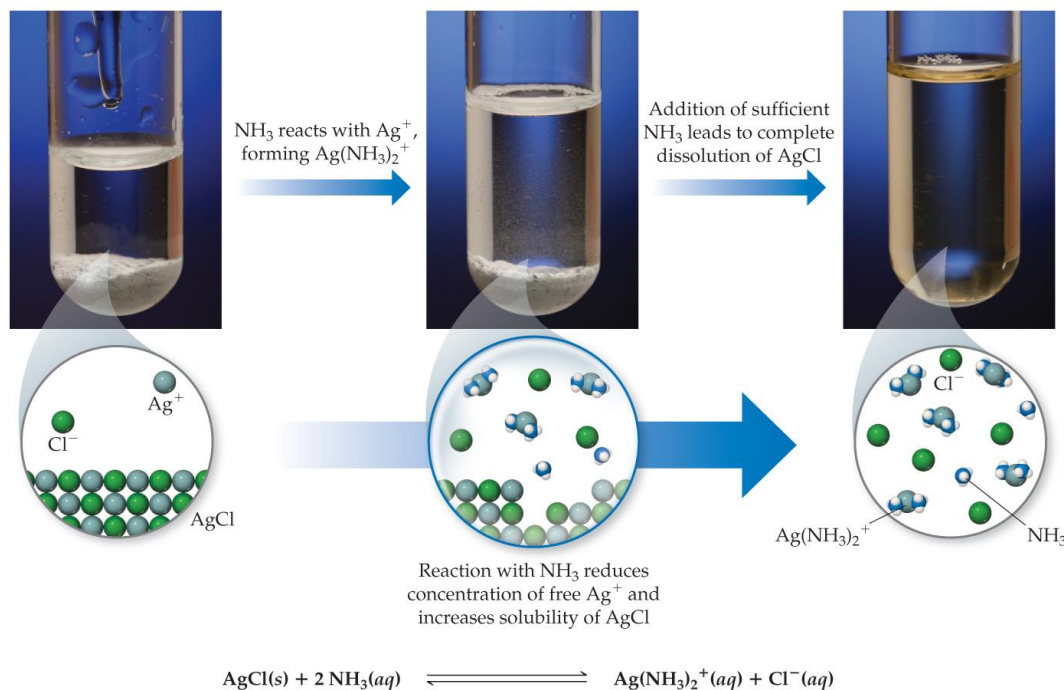
- Metal ions can act as Lewis acids and form **complex ions** with Lewis bases in the solvent.
- The formation of these complex ions increases the solubility of these salts.

Table 17.1 Formation Constants for Some Metal Complex Ions in Water at 25 °C

Complex Ion	K_f	Equilibrium Equation
$\text{Ag}(\text{NH}_3)_2^+$	1.7×10^7	$\text{Ag}^+(aq) + 2 \text{NH}_3(aq) \rightleftharpoons \text{Ag}(\text{NH}_3)_2^+(aq)$
$\text{Ag}(\text{CN})_2^-$	1×10^{21}	$\text{Ag}^+(aq) + 2 \text{CN}^-(aq) \rightleftharpoons \text{Ag}(\text{CN})_2^-(aq)$
$\text{Ag}(\text{S}_2\text{O}_3)_2^{3-}$	2.9×10^{13}	$\text{Ag}^+(aq) + 2 \text{S}_2\text{O}_3^{2-}(aq) \rightleftharpoons \text{Ag}(\text{S}_2\text{O}_3)_2^{3-}(aq)$
CdBr_4^{2-}	5×10^3	$\text{Cd}^{2+}(aq) + 4 \text{Br}^-(aq) \rightleftharpoons \text{CdBr}_4^{2-}(aq)$
$\text{Cr}(\text{OH})_4^-$	8×10^{29}	$\text{Cr}^{3+}(aq) + 4 \text{OH}^-(aq) \rightleftharpoons \text{Cr}(\text{OH})_4^-(aq)$
$\text{Co}(\text{SCN})_4^{2-}$	1×10^3	$\text{Co}^{2+}(aq) + 4 \text{SCN}^-(aq) \rightleftharpoons \text{Co}(\text{SCN})_4^{2-}(aq)$
$\text{Cu}(\text{NH}_3)_4^{2+}$	5×10^{12}	$\text{Cu}^{2+}(aq) + 4 \text{NH}_3(aq) \rightleftharpoons \text{Cu}(\text{NH}_3)_4^{2+}(aq)$
$\text{Cu}(\text{CN})_4^{2-}$	1×10^{25}	$\text{Cu}^{2+}(aq) + 4 \text{CN}^-(aq) \rightleftharpoons \text{Cu}(\text{CN})_4^{2-}(aq)$
$\text{Ni}(\text{NH}_3)_6^{2+}$	1.2×10^9	$\text{Ni}^{2+}(aq) + 6 \text{NH}_3(aq) \rightleftharpoons \text{Ni}(\text{NH}_3)_6^{2+}(aq)$
$\text{Fe}(\text{CN})_6^{4-}$	1×10^{35}	$\text{Fe}^{2+}(aq) + 6 \text{CN}^-(aq) \rightleftharpoons \text{Fe}(\text{CN})_6^{4-}(aq)$
$\text{Fe}(\text{CN})_6^{3-}$	1×10^{42}	$\text{Fe}^{3+}(aq) + 6 \text{CN}^-(aq) \rightleftharpoons \text{Fe}(\text{CN})_6^{3-}(aq)$



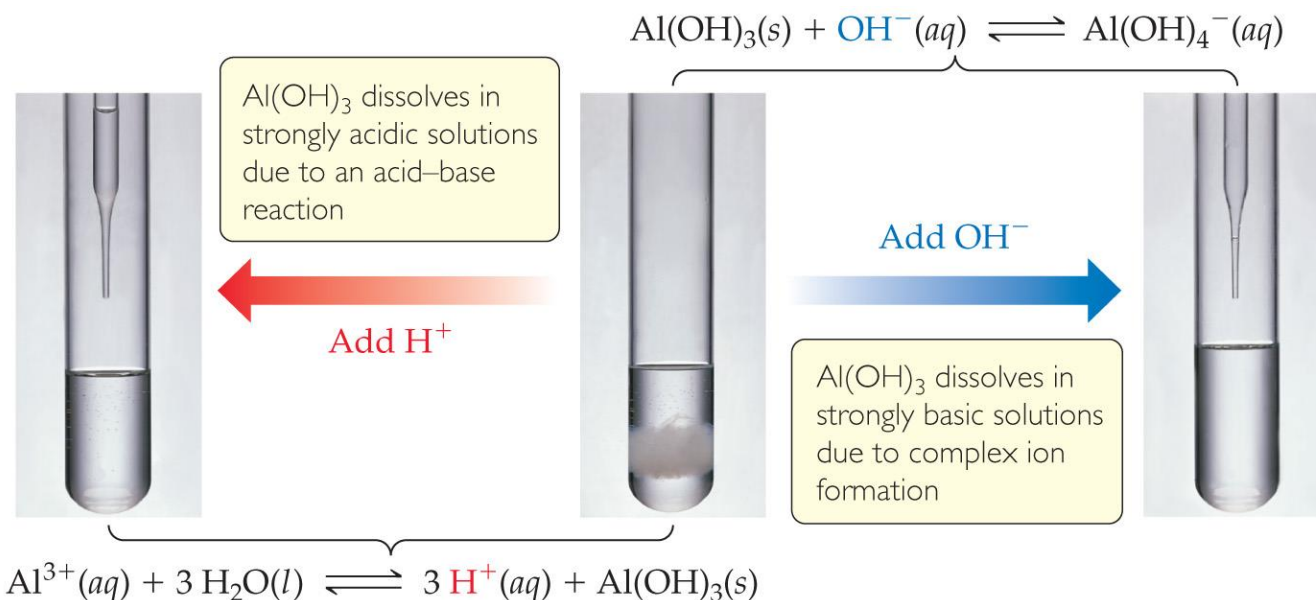
How Complex Ion Formation Affects Solubility



- Silver chloride is insoluble. It has a K_{sp} of 1.6×10^{-10} .
- In the presence of NH₃, the solubility greatly increases because Ag⁺ will form complex ions with NH₃.

Amphoterism and Solubility

- **Amphoteric** oxides and hydroxides are soluble in strong acids or base, because they can act either as acids or bases.
- Examples are oxides and hydroxides of Al^{3+} , Cr^{3+} , Zn^{2+} , and Sn^{2+} .



Will a Precipitate Form?

- To decide, we calculate the reaction quotient, Q , and compare it to the solubility product constant, K_{sp} .
 - If $Q = K_{sp}$, the system is at equilibrium and the solution is saturated.
 - If $Q < K_{sp}$, more solid can dissolve, so no precipitate forms.
 - If $Q > K_{sp}$, a precipitate will form.



Tooth Decay and Fluoridation



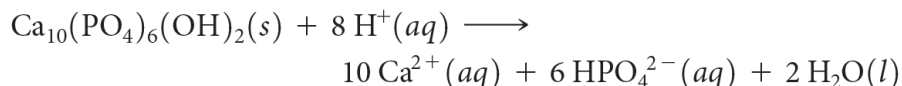
Amazon.com : Colgate Cavity...
www.amazon.com





Tooth Decay and Fluoridation

Tooth enamel consists mainly of the mineral hydroxyapatite, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, the hardest substance in the body. Tooth cavities form when acids dissolve tooth enamel:

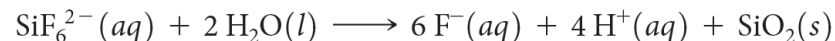


The Ca^{2+} and HPO_4^{2-} ions diffuse out of the enamel and are washed away by saliva. The acids that attack the hydroxyapatite are formed by the action of bacteria on sugars and other carbohydrates present in the plaque adhering to the teeth.

Fluoride ion, which is added to municipal water systems and toothpastes, can react with hydroxyapatite to form fluoroapatite,

$\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$. This mineral, in which F^- has replaced OH^- , is much more resistant to attack by acids because the fluoride ion is a much weaker Brønsted–Lowry base than the hydroxide ion.

The usual concentration of F^- in municipal water systems is 1 mg/L (1 ppm). The compound added may be NaF or Na_2SiF_6 . The silicon-fluorine anion reacts with water to release fluoride ions:



About 80% of all toothpastes now sold in the United States contain fluoride compounds, usually at the level of 0.1% fluoride by mass. The most common compounds in toothpastes are sodium fluoride (NaF), sodium monofluorophosphate ($\text{Na}_2\text{PO}_3\text{F}$), and stannous fluoride (SnF_2).

Related Exercises: 17.100, 17.118



SAMPLE EXERCISE 17.16 Predicting Whether a Precipitate Forms

Does a precipitate form when 0.10 L of $8.0 \times 10^{-3} \text{ M Pb(NO}_3)_2$ is added to 0.40 L of $5.0 \times 10^{-3} \text{ M Na}_2\text{SO}_4$?

$$K_{sp} = 1.6 \times 10^{-10} \text{ for PbSO}_4$$

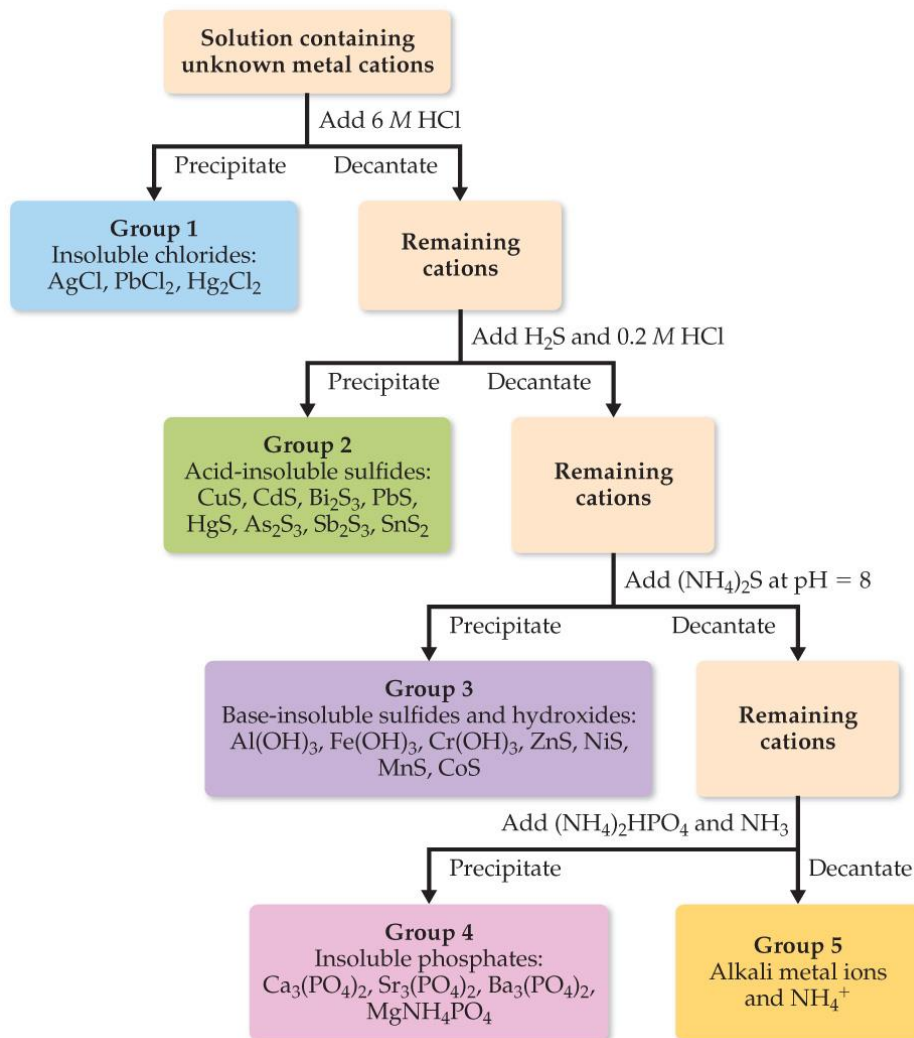


SAMPLE EXERCISE 17.17 Selective Precipitation

A solution contains $1.0 \times 10^{-2} M \text{Ag}^+$ and $2.0 \times 10^{-2} M \text{Pb}^{2+}$. When Cl^- is added, both AgCl ($K_{sp} = 1.8 \times 10^{-10}$) and PbCl_2 ($K_{sp} = 1.7 \times 10^{-5}$) can precipitate. What concentration of Cl^- is necessary to begin the precipitation of each salt? Which salt precipitates first?



Selective Precipitation of Ions



One can use differences in solubilities of salts to separate ions in a mixture. This has been used for qualitative analysis of the presence of ions in a solution.

